

UX Evaluation: Goals, Metrics, and Targets

Highlights

- Concepts of UX goals, metrics, and targets
- UX target tables
 - Work role and user classes
 - UX goals
 - UX measures
 - Measuring instruments
 - UX metrics
 - Level setting
- Practical tips and cautions
- Rapid goals, metrics, and targets

20.1 INTRODUCTION

20.1.1 You Are Here

We begin each process chapter with a “You are here” picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 20.1). In this chapter, we establish quantitative operational targets for measures of user experience to assess the level of success in your designs so that you know when you can move on to the next iteration.

UX goals, metrics, and targets help build scaffolding to support planning for evaluation that will reveal problems with user performance and emotional satisfaction. If used, UX goals, metrics, and targets are set up early as part of preparation for evaluation.

20.1.2 Project Context for UX Metrics and Targets

In the early stages, evaluation usually focuses on qualitative data for finding UX problems. Later, you can use quantitative metrics to see how you are doing. Or, especially in an agile environment, you may wish to forego UX metrics and targets altogether.

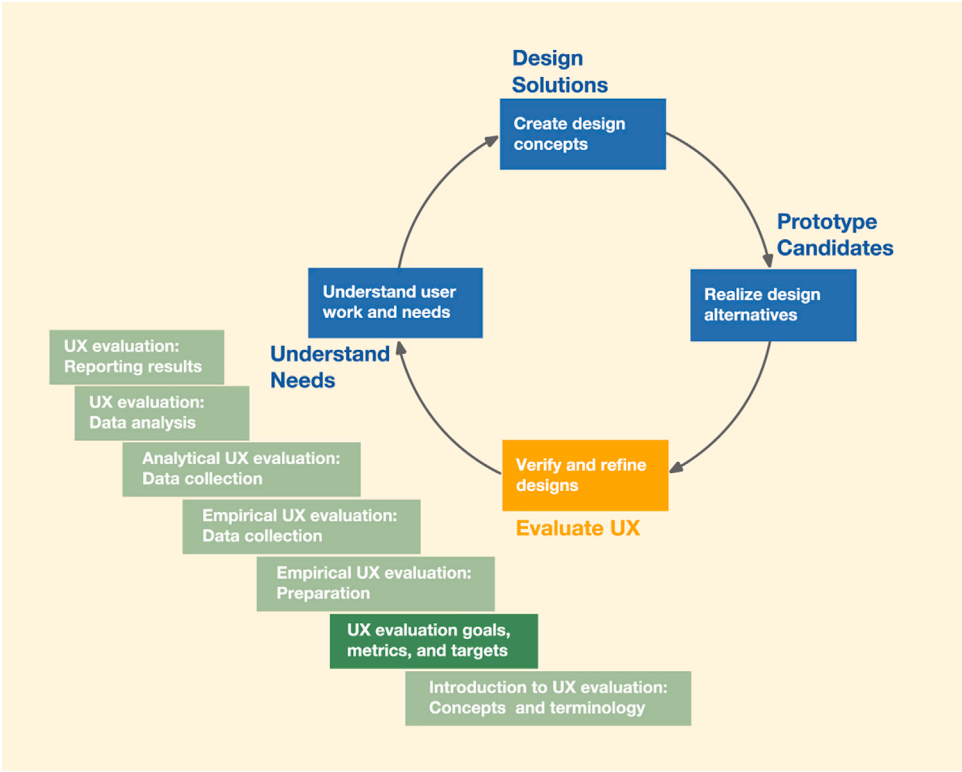


Fig. 20.1
You are here in the chapter on UX evaluation goals, metrics, and targets within the evaluate UX activity in the context of the overall Wheel UX lifecycle.

Experienced UX designers can usually know which parts of the design need improvement just by the qualitative results of the first round of evaluation. In such cases, quantitative UX metrics and targets (Sections 20.6 and 20.8) may not be worth the investment, but benchmark tasks (Section 20.5.1) are still useful as vehicles for driving rapid qualitative evaluation of usability, user satisfaction, and enjoyment. In any case, if you need to establish quantitative metrics to guide your UX evaluation, then this chapter shows you how.

20.1.3 UX Target Tables

Through years of working with real-world UX professionals and doing our own UX evaluations, we have refined the concept of a UX target table (Table 20.1) from the original concept of a usability specification table (Whiteside, Bennett, & Holtzblatt, 1988). A spreadsheet is an obvious way to implement these tables.

20.2 WORK ROLE AND USER CLASSES

The first column in Table 20.1 is for the work role (see Section 8.2) and related user class (the user population intended to take on this work role [see

Table 20.1

Our UX target table. For convenience, a single row is called a UX target. Adapted from Whiteside, Bennett, and Holtzblatt (1988).

Work Role: User Class	UX Goal	UX Measure	Measuring Instrument	UX Metric	Baseline Level	Target Level	Observed Results
← UX target →							

Section 8.2.5]) to which a UX target in a given row in the table applies. The next two columns in Table 20.1 are for the related UX goal and the associated UX measure, which go together because each UX measure is aimed at supporting a UX goal and is specified with respect to a work role and user class combination. Next, you will see where you get the information for these three columns.

Example: Work Role and User Class for the Ticket Kiosk System

For the Ticket Kiosk System, we begin by focusing on the user work role of the ticket buyer. For this work role, user classes could include a casual town resident user from Middleburg or a student user from Middleburg University. In this example, we feature the casual town user (Table 20.2).

Table 20.2

Choosing a work role and a user class for a UX target

Work Role: User Class	UX Goal	UX Measure	Measuring Instrument	UX Metric	Baseline Level	Target Level	Observed Results
Ticket buyer: Town resident, casual new user, for occasional personal use							

20.3 UX GOALS

UX goals are high-level objectives for a UX design that are stated in terms of user experience aspirations. UX goals can be driven by business goals and reflect real use of a product and identify what is important to an organization, its customers, and its users.

Examples of UX goals include ease of use for all users, ease of remembering for intermittent users, power performance for experts, avoiding errors for safety-critical systems, high customer satisfaction, walk-up-and-use learnability for new users, among others.

Example: UX Goals for the Ticket Kiosk System

From our UX research data, we can define the primary high-level UX goals for the ticket buyer to include:

- Fast-and-easy walk-up-and-use user experience, with no user training
- Fast learning so new user performance after limited experience is on par with that of an experienced user
- High customer satisfaction leading to a high rate of repeat customers

Other possibilities include:

- High learnability for more advanced tasks
- Draw, engagement, and attraction
- Low error rate for completing transactions, especially in the interaction for payment

A UX target table entry for the UX goal of “fast-and-easy walk-up-and-use user experience” is straightforward. This goal refers to the ability of a typical occasional user to do at least the basic tasks on the first try, without training or manuals. We see the beginnings of a UX target in [Table 20.3](#).

Table 20.3
Choosing a UX goal for a UX target

Work Role: User Class	UX Goal	UX Measure	Measuring Instrument	UX Metric	Baseline Level	Target Level	Observed Results
Ticket buyer: Town resident, casual new user, for occasional personal use	Walk-up ease of use for new user						

Exercise 20.1 Identify UX Evaluation Goals for Your Product or System

Goal: Gain experience in stating UX goals (see [Section 5.6.1.2](#)).
Activities: Review the work products of your UX research data analysis (see [Chapter 7](#)) for the system of your choice, noting user or customer concerns relating to UX goals.

Deliverables: A short list of UX goals for one user class of the system of your choice.

Schedule: Take approximately 30 minutes for this rather easy exercise.

20.4 UX MEASURES

Within a UX target, the UX measure is the general user experience characteristic to be measured with respect to usage of your UX design. Example UX measures include initial task performance, long-term task performance, first impression, and initial satisfaction. Initial performance refers to a user’s performance during the very first use (somewhere between the first few minutes and the first few hours, depending on the complexity of the system). Initial performance and first impressions are appropriate UX measures for virtually every UX design.

Example: UX Measures for the Ticket Kiosk System

For the walk-up ease-of-use goal of our casual new user, we begin simply with initial performance and add a new goal of initial satisfaction with first impression as the measure. Each UX goal and corresponding UX measure will appear in separate UX targets in the table, with the work role and user class repeated, as in [Table 20.4](#).

Table 20.4
Choosing initial performance and first impression as UX measures

Work Role: User Class	UX Goal	UX Measure	Measuring Instrument	UX Metric	Baseline Level	Target Level	Observed Results
Ticket buyer: Town resident, casual new user, for occasional personal use	Walk-up ease of use for new user	Initial user performance					
Ticket buyer: Town resident, casual new user, for occasional personal use	Initial customer satisfaction	First impression					

20.5 MEASURING INSTRUMENTS

Within a UX target, a measuring instrument is the means of generating values for the particular UX measure; it is the vehicle that creates the values that are measured.

20.5.1 Measuring Instruments: Benchmark Tasks

A benchmark task is a representative task that you will have user participants attempt to accomplish in evaluation while you observe their performance and behavior. As such, a benchmark task is a standardized task that can be used to compare performance among different users and across different design versions.

Although you can get creative in choosing your measuring instruments, objective measures are commonly associated with a benchmark task (e.g., a time-on-task measure as timed on a stopwatch, or an error rate measure made by counting user errors). Subjective measures are commonly associated with a user questionnaire (e.g., measuring the average user rating scale scores for a specific set of questions).

20.5.1.1 *Selecting benchmark tasks*

Choose realistic tasks expected to be used by each user class of a work role across the system. To get the best coverage for your evaluation investment, your choices should represent the cross section of real tasks with respect to frequency of performance and criticality to goals of users of the envisioned product. Benchmark tasks are also selected to evaluate new features, edge cases, and business- or mission-critical tasks. Although some of these tasks may not be performed frequently, getting them wrong could cause serious consequences.

Example: Initial Benchmark Task Choice for the Ticket Kiosk System

For our Ticket Kiosk System, you could begin by finding a movie that is currently playing, then follow by searching for and reserving available tickets for a movie that is to be showing 20 days from now. Match the benchmark task to the UX measure. For example, if the UX measure is “initial user performance,” then the task should be among those a first-time user realistically would face.

Use tasks in realistic combinations to evaluate task flow. For example, in the Ticket Kiosk System, you may wish to measure user performance on the task thread of searching for an event and then buying tickets with seat selection for that event.

Example: Benchmark Task for the Ticket Kiosk System

As another example, a benchmark task may require users to buy four tickets for a concert under a total of \$200, while showing tickets (in the prototype design being evaluated) in this price range for the upcoming few days as sold out. This

would force users to perform the task of searching through other future concert days, looking for the first available day with tickets in this price range.

Example: Benchmark Task as Measuring Instrument

For the Ticket Kiosk System, the first UX target in Table 20.5 contains an objective UX measure for “Initial user performance.” An obvious choice for the corresponding measuring instrument is a benchmark task. Here we need a simple and frequently used task that can be done in a short time by a casual new user in a walk-up ease-of-use situation. An appropriate benchmark task would involve buying tickets to an event. Here is a possible description to give the user participant:

“BT1: Go to the Ticket Kiosk System and buy three tickets for the Monster Truck Pull on February 28 at 7 p.m. Get three seats together as close to the front as possible. Pay with a major credit card.”

In Table 20.5, we add this as the measuring instrument for the first UX target.

Table 20.5
Entering BT1 (buy three tickets to the Monster Truck Pull) benchmark task as a measuring instrument for “initial user performance” UX measure for the first UX target

Work Role: User Class	UX Goal	UX Measure	Measuring Instrument	UX Metric	Baseline Level	Target Level	Observed Results
Ticket buyer: Town resident, casual new user, for occasional personal use	Walk-up ease of use for new user	Initial user performance	BT1: Buy Monster Truck Pull tickets				
Ticket buyer: Town resident, casual new user, for occasional personal use	Initial customer satisfaction	First impression					

20.5.1.2 Crafting benchmark task descriptions

General guidelines for creating effective benchmark tasks include the following:

- Remove any ambiguities with clear, precise, specific, and repeatable instructions.
- Tell the user what task to do, but not how to do it.

Example: Instructional Wording in a Benchmark Task

To do: “Buy two student tickets for available adjacent seats as close to the stage as possible for the upcoming Iris Dement concert and pay with a credit card.”

Not to do: “Click on the Special Events button on the home screen; then select More at the bottom of the screen. Select the Iris Dement concert and click on Seating Options...”

- Avoid using words in benchmark tasks that appear specifically in the UX design. For example, don’t say “Find such and such” when there is a button in the design labeled “Find.”
- Use work context and usage-centered wording, not system-oriented wording. For example, “Find information about xyz” is better than “Submit query about xyz.”
- Keep some mystery in it for the user. Avoid being too specific about what the users will see or the parameters they will encounter. Remember, most users do not think in precise and design-oriented terms.
- Ensure ecological validity (Thomas & Kellogg, 1989); write benchmark tasks to match the user’s real work context.
- Make the benchmark task doable in a reasonable amount of time—not too long for a UX evaluation session but not too short to be meaningful.

Example: Ecological Validity in Benchmark Tasks for the Ticket Kiosk System

To evaluate use of the Ticket Kiosk System to manage the work activity of ticket buying, you can make good use of physical prototypes and representative locations to enhance ecological validity. For example, you could mount a real touchscreen device inside a physical cardboard or wooden mockup of a kiosk structure. When placed in a hallway of a relatively busy work area, this will attract lots of interest and curious people who will want to try it out (Gould et al., 1987).

20.5.2 Measuring Instruments: User Satisfaction Questionnaires

As a measuring instrument for a subjective UX measure, a questionnaire targeting various UX design features can be used to determine a user’s satisfaction with the UX design. Measuring a user’s satisfaction provides a subjective but still quantitative UX metric for the related UX measure. In the following examples, we use the Questionnaire for User Interface Satisfaction (QUIS) (Chin, Diehl, & Norman, 1988) (see [Section 22.2.2.1](#)).

Example: Questionnaire as Measuring Instrument for the Ticket Kiosk System

You can specify a particular questionnaire be administered following the initial tasks, stipulating it as the measuring instrument in the second UX target of the growing UX target table (Table 20.6).

Table 20.6
Choosing a questionnaire as the measuring instrument for first impression UX measure

Work Role: User Class	UX Goal	UX Measure	Measuring Instrument	UX Metric	Baseline Level	Target Level	Observed Results
Ticket buyer: Town resident, casual new user, for occasional personal use	Walk-up ease of use for new user	Initial user performance	BT1: Buy Monster Truck Pull tickets				
Ticket buyer: Town resident, casual new user, for occasional personal use	Initial customer satisfaction	First impression	First five questions in the Questionnaire for User Interface Satisfaction				

20.6 UX METRICS

A UX metric describes a specific characteristic for which you are getting a value. It is the name of the value you get for a given UX target. For example, in Table 20.7, the “user ratings averaged over the first five questions” is the metric for measuring first impressions. There can be more than one metric for a given measure.

Probably the most common UX metrics are objective, performance oriented, and taken while the participant is carrying out a benchmark task. Other UX metrics can be subjective, based on a rating or score computed from questionnaire results. Typical objective UX metrics include time to complete a task¹ and number of errors made by the user.

¹Although the time on task often makes a useful UX metric, it clearly is not appropriate in some cases. For example, if the task performance time is affected by factors beyond the user’s control, then time on task is not a good measure of user performance. This exception includes cases of long and/or unpredictable communication and response-time delays, such as might be experienced in some website usage.

Table 20.7
Choosing UX metrics for UX measures

Work Role: User Class	UX goal	UX Measure	Measuring Instrument	UX Metric	Baseline Level	Target Level	Observed Results
Ticket buyer: Town resident, casual new user, for occasional personal use	Walk-up ease of use for new user	Initial user performance	BT1: Buy Monster Truck Pull tickets	Average time on task (over all users)			
Ticket buyer: Town resident, casual new user, for occasional personal use	Initial customer satisfaction	First impression	First five questions in the Questionnaire for User Interface Satisfaction (QUIS)	Average rating across users and across first five questions of QUIS			

Typically, subjective UX metrics will represent a numeric outcome from a questionnaire, usually based on simple arithmetic statistical measures, such as the numeric average. Remember that you are going only for an engineering indicator of user experience, not for statistical significance.

Finally, be sure you match up your UX measures, measuring instruments, and metrics to make sense in a UX target. For example, if you plan to use a questionnaire in a UX target, avoid naming the UX measure “Initial Performance.” A questionnaire does not measure performance; it measures user satisfaction or opinion.

Example: UX Metrics for the Ticket Kiosk System

For the initial performance UX measure in the first UX target of Table 20.7, as already discussed in Section 20.4, the average length of time to buy a particular special event ticket is an appropriate value to measure. We specify this by adding “average time on task, averaged over all users in the evaluation” as the UX metric in the first UX target of Table 20.7.

As a different objective performance measure, you could measure the average number of errors a user makes while buying a concert ticket (see Table 20.9, later). For this target, the measuring instrument is BT3, and the UX metric is the average number of errors. You will often want to measure multiple metrics during a participant’s single performance of the same

benchmark task. A participant does not, for example, need to perform one “buy ticket” task while you time performance and then do a different (or repeat the same) “buy ticket” task while you count errors.

20.7 BASELINE LEVEL

The baseline level is the hypothetical benchmark level of the UX metric; it is the talking point level against which other levels are compared. For this, evaluators often use a level that has been measured for the current version of the system (automated or manual). Sometimes setting baseline levels can be a challenge. The answer requires determining what level of user performance and user experience the system currently supports.

Example: Baseline Level Values for the Ticket Kiosk System

To determine the values for the first two UX baseline levels for the Ticket Kiosk System, we can have someone perform the benchmark tasks using the Middleburg University Ticket Transaction Service (MUTTS) (the old existing system) ticket counter. The resulting time on task may be quite different from what you expect users will be able to achieve using our new system, but at least it is something for comparison. Measuring a baseline level helps ensure that the UX metric is, in fact, measurable.

Suppose we find that buying a ticket for the Monster Truck Pull event at the current MUTTS counter takes about 3 minutes. Then this is a plausible baseline level for the first UX target in [Table 20.8](#). Because most people are already experienced with ticket offices, this value is not really for initial performance, but it gives some idea for that value.

To establish a baseline value for the first impression UX measure in the second UX target, we could administer the questionnaire to some users of MUTTS. Let us say we have done that, and our average score is 6 of 10 for the first impression UX measure (see [Table 20.8](#)).

20.8 TARGET LEVEL

The target-level specification is an operationally defined criterion for success of the expected user experience. It is a quantitative statement of an aimed-at or hoped-for value for a UX metric. The target level for a UX metric is the minimum value indicating attainment of user experience success. So, a target level is usually set to be an improvement over the corresponding baseline level. Remember that the target levels in a table refer to the average overall

Table 20.8
Setting baseline levels for UX measures

Work Role: User Class	UX goal	UX Measure	Measuring Instrument	UX Metric	Baseline Level	Target Level	Observed Results
Ticket buyer: Town resident, casual new user, for occasional personal use	Walk-up ease of use for new user	Initial user performance	BT1: Buy Monster Truck Pull tickets	Average time on task (over all users)	3 minutes, measured at the Middleburg University Ticket Transaction Service (MUTTS) counter		
Ticket buyer: Town resident, casual new user, for occasional personal use	Initial customer satisfaction	First impression	First five questions in the Questionnaire for User Interface Satisfaction (QUIS)	Average rating across users and across the first five questions of QUIS	6/10, from MUTTS users		

participants of the corresponding measure, meaning that the levels shown do not have to be achieved by every participant in every evaluation session.

Because passing the user experience test means meeting all your target levels simultaneously, you have to ensure that the target levels for all UX measures in the entire table must be simultaneously attainable.

Obviously, level values are often best guesses; however, with practice, UX people become quite skilled at establishing reasonable and credible target levels and setting reasonable values. As an example, for an initial performance measure, an experienced UX practitioner will set target levels that allow enough time for unfamiliar users to read menus and labels, think a bit, and look around each screen to get their bearings.

Example: Target-level Values for the Ticket Kiosk System

For the first initial performance UX measure, we set the first target level at 2.5 minutes, a reasonable choice with respect to our baseline level of 3 minutes. We enter this value into the “Target level” column for the first UX target in Table 20.9.

Given that our baseline level for the second target is based on MUTTS users, we should aspire to significant improvement and enter, say, a level of 8 as the target level for this questionnaire rating.

Table 20.9

Setting target levels for UX metrics

Work Role: User Class	UX Goal	UX Measure	Measuring Instrument	UX Metric	Baseline Level	Target Level	Observed Results
Ticket buyer: Town resident, casual new user, for occasional personal use	Walk-up ease of use	Initial user performance	BT1: Buy Monster Truck Pull tickets	Average time on task (over all users)	3 minutes, measured at the Middleburg University Ticket Transaction Service (MUTTS) ticket counter	2.5 minutes	
Ticket buyer: Town resident, casual new user, for occasional personal use	Initial customer satisfaction	First impression	First five questions in the Questionnaire for User Interface Satisfaction (QUIS)	Average rating across users and across the first five questions of QUIS	6/10, from MUTTS users	8/10	
Ticket buyer: Frequent music patron	Accuracy	Experienced usage error rate	BT3: Buy tickets for Schubert String Quartet concert at Moss Arts Center on campus	Average number of errors	<1	<1	
Casual public ticket buyer	Walk-up ease of use for new user	Initial user performance	BT4: Buy Monster Truck Pull tickets	Average time on task	5 minutes (online system)	2.5 minutes	
Casual public ticket buyer	Walk-up ease of use for new user	Initial user performance	BT4: Buy Monster Truck Pull tickets	Average number of errors	<1	<1	
Casual public ticket buyer	Walk-up ease of use for user with a little experience	Just following initial performance	BT5: Buy <i>Top Gun: Maverick</i> movie tickets	Average time on task	5 minutes (including review)	2 minutes	

Considering the target level for the third target, with a baseline level of less than one error for the concert ticket purchasing task, it would be tempting to set the target level at zero because users should not be making errors on such as easy task. However, a target level of zero, being an average over all user participants in the evaluation, does not allow for anyone ever to commit an error. So, let us set the level as less than one, entered as the target level for error rates into the third UX target of [Table 20.9](#). Beyond the third UX target in [Table 20.9](#), we have added a few other UX targets as examples.

20.9 OBSERVED RESULTS

The final column in [Table 20.9](#) is for observed results, a space reserved for recording values measured while observing users performing the prescribed tasks during formative evaluation sessions. As part of the UX target table, this column affords direct comparisons between specified levels and results of testing.

Because you typically will have more than one user from which observed results are obtained, you can either record multiple values in a single observed results column or, if desired, add more columns for observed results and use this column for the average of the observed values. If you maintain your UX target tables in spreadsheets, as we recommend, it is easier to manage observed data and results later in UX evaluation analysis (see [Chapter 24](#)).

Exercise 20.2 Create Benchmark Tasks and UX Targets for Your System

Goal: To gain experience in writing effective benchmark tasks and measurable UX targets (see [Section 5.6.1.2](#)).

Activities: We have shown you a rather complete set of examples of benchmark tasks and UX targets for the Ticket Kiosk System. Your job is to do something similar for the system of your choice.

- Begin by identifying which work roles and user classes you are targeting in evaluation (brief description is enough).
- Create three or more UX targets as UX table entries (rows), including your values for each column. Have at least two UX targets based on a benchmark task and at least one based on a questionnaire.
- Create and write a set of approximately three benchmark tasks to go with the UX targets in the table. Do not make the tasks too easy.

- Make tasks increasingly complex.
- Include some navigation.
- Create tasks that you can later implement in your low-fidelity rapid prototype.
- Include the questionnaire question numbers in the measuring instrument column of the appropriate UX target.

Cautions and hints:

- Do not spend much time on design in this exercise; here we are focusing on UX goals and metrics.
- Do not plan to give users any training.

Deliverables:

- Two user benchmark tasks, each on a separate sheet of paper.
- Three or more UX targets entered into a blank UX target table on your laptop or on paper.
- If you are doing this exercise in a classroom environment, finish up by reading your benchmark tasks to the class for critique and discussion.

Schedule: Work efficiently and complete in about 90 minutes.

20.10 SEGUE

In this chapter, we discussed setting the goals, metrics, and targets for UX evaluation. In the next chapter, we get into preparing for an empirical evaluation study.